COE817 Lab 1
Programming with Sockets in Java

Objectives

- In this lab, you will familiar yourself with Java socket programming.
- You will work on an implementation of the Vigenère Cipher, the system for encoding and decoding text messages that was discussed in class.
- Two students will work as a group. You will program with Java. Netbeans is the IDE.
- Duration: two weeks.

Preparing for Lab 1:

Familiar yourself with the following Java socket tutorials.

(1) Echo: an example of a server handles one client

https://docs.oracle.com/javase/tutorial/networking/sockets/definition.html

https://docs.oracle.com/javase/tutorial/networking/sockets/readingWriting.html

The links of EchoClient.java and EchoServer.java are in the tutorial.

(2) Knock Knock jokes: another example of a server handles one client

https://docs.oracle.com/javase/tutorial/networking/sockets/clientServer.html

The links of KnockKnockClient.java, KnockKnockServer.java and KnockKnockProtocol.java are in the tutorial.

(3) Knock Knock jokes: an example of a server handles multiple clients simultaneously

https://docs.oracle.com/javase/tutorial/networking/sockets/clientServer.html

The links of KKMultiServer.java and KKMultiServerThread.java are in the tutorial

Please note that your clients and server will be run on a same machine. In order to run both client and server applications on the same host you should bind your server socket to localhost (you can actually write "localhost" or 127.0. 0.1). Localhost always refers to the computer you work on.

Siri text chat

- Design Java socket programs to simulate an online Siri text chat system.
- The Siri text chat system is a client-server application. One client or multiple clients send questions to a Siri server. The server responds the clients with answer messages

Sample Q/A chat messages between Siri sever and clients could be as follows:

Q: Who created you?
A: I was created by Apple.

Q: What does Siri mean?
A: victory and beautiful

Q: Are you a robot?
A: I am a virtual assistant.

You are free to add more Q/A messages in your programs.

- A client process contacts the server by connecting to the port number you choose on the server. The server process accepts the connection request and then handles the question from the client.
- When server handles multiple clients, the server process need to create thread process to handle clients using new sockets.

Your Task

Project 1: server handles one client

(1) In Netbeans, create your lab 1 directory, save your program as "**Project1**" on your lab1 directory.

(2) Write Java socket programs to make a client to be able to chat with a Siri server. In this project, you only need to make a server handle one client.

(3) Encode and decode chat Q/A messages between Siri server and client using Vigenère cipher. We assume that the Vigenère key is "TMU".

- When you run your programs, the client’s question needs to be encrypted with the Vigenère cipher.
- After receive client’s encrypted message, the server will display it, decrypt it and also show the decrypted message.
- Similarly, when server responses answer to the client, the server need to encrypt it before sending it to the client
- After the client receives the encrypted answer from the server, the client will display the received encrypted answer, decrypt it and show the decrypted answer.

Project 2: server handles multiple clients simultaneously

- In Netbeans, create your lab 1 directory, save your program as "**Project2**" on your lab1 directory.
- Write Java socket programs to make multiple clients chat with a Siri server simultaneously.

What you needed to demonstrate to your TA

1. For project 1, show your encoding and decoding results.
2. For project 2, show different clients chat with the Siri server.

3. Go over your programs, locate and explain the parts of the code that deal with the following implementations:

- How could your server program handle multiple clients?
- Where did you encrypt chat messages in your program?
- Where did you decrypt chat messages in your program?

Submitting your lab

You must submit your source codes of project 1 and project 2 to D2L lab1 submission folder at least 24 hours prior to the start of your scheduled lab period for Lab 2.